

Introduction to Biochemistry

Introduction to Biochemistry

- **Biochemistry is the study of molecular basis of life.**
- **It is important to know biochemistry so as to understand and maintain normal health and to know how to treat disease**

Biochemistry

- **Biochemistry is the science concerned with the study chemical constituents of living cells (biomolecules), the basic units of life...what is life?**

Biomolecules

1- carbohydrates

2- proteins

3- lipids

4- nucleic acids

5- vitamins

6- water

7- minerals

Biochemistry

- **Study of Biomolecules:**
 - * **Structures**
 - * **Functions**
 - * **Transformations & energetic changes associated with them in the course of the activities of living organisms.**

Biochemistry

- **The major objective of biochemistry is the complete understanding at the molecular level of all processes associated with living cells.**

- ◎ Life is based on chemical reactions in living cells.
- ◎ We want to know these reactions, the reacting molecules, their nature, properties, in which form we get them from the environment, how do they gain access to our cells, their effect on cell growth and cell division and differentiation...
- ◎ all activities of living organisms ..thinking..emotions..

Biochemistry

- **Explanation of muscle contraction and movement at the molecular level
..actin and myosin are proteins**
- **Another objective of biochemistry is to attempt to understand how life began (evolution)**



41A



41



Biochemistry

- **What happens if we have imbalance or disturbance of normal biomolecules..... diseases**

Nutrients are chemical substances essential for growth and maintenance of normal cell.

About 45-50 chemical substances are known to be essential or required by the human for maintenance of health.





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Glycogen Storage Disease type I:
clinical, biochemical and genetic aspects,
and implications for treatment and follow-up

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Biochemistry

- ◎ Biochemistry is involved in all life sciences physiology, microbiology, veterinary, pharmacology, agriculture, oncology, pathology, immunology, medicine....
- ◎ Nutrition, Cell biology, molecular biology, and molecular genetics, recombinant DNA technology are considered as branches or developments within biochemistry

- **Biochemical techniques improved knowledge in many aspects of physiology, immunology, pharmacology toxicology, cancer and pathology.**
- **Improved life stock and crops**
- **Biological and chemical warfare**

The story of Dolly...

On 5th of July 1996 cloning was revolutionized when Ian Wilmut and his colleagues successfully cloned a sheep named Dolly at the Roslin Institute in Scotland.

She was cloned from and adult cell.
 *How????*

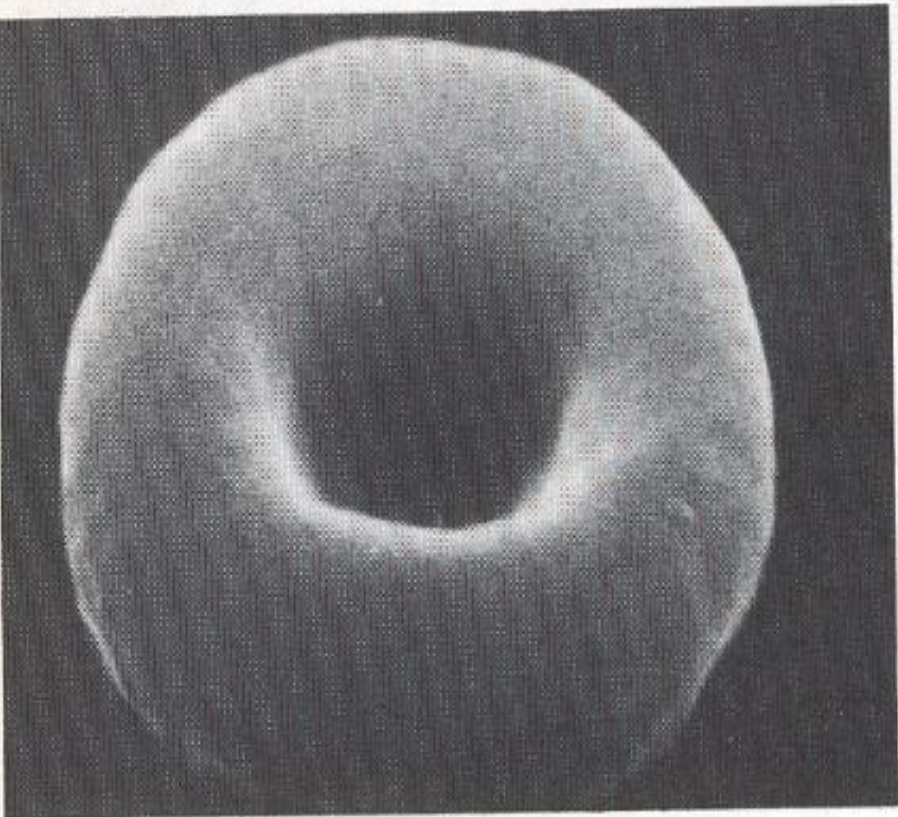
It took 276 attempts

Dolly died in early 2002, at the age of 6, so her cells are claimed to 12 years old

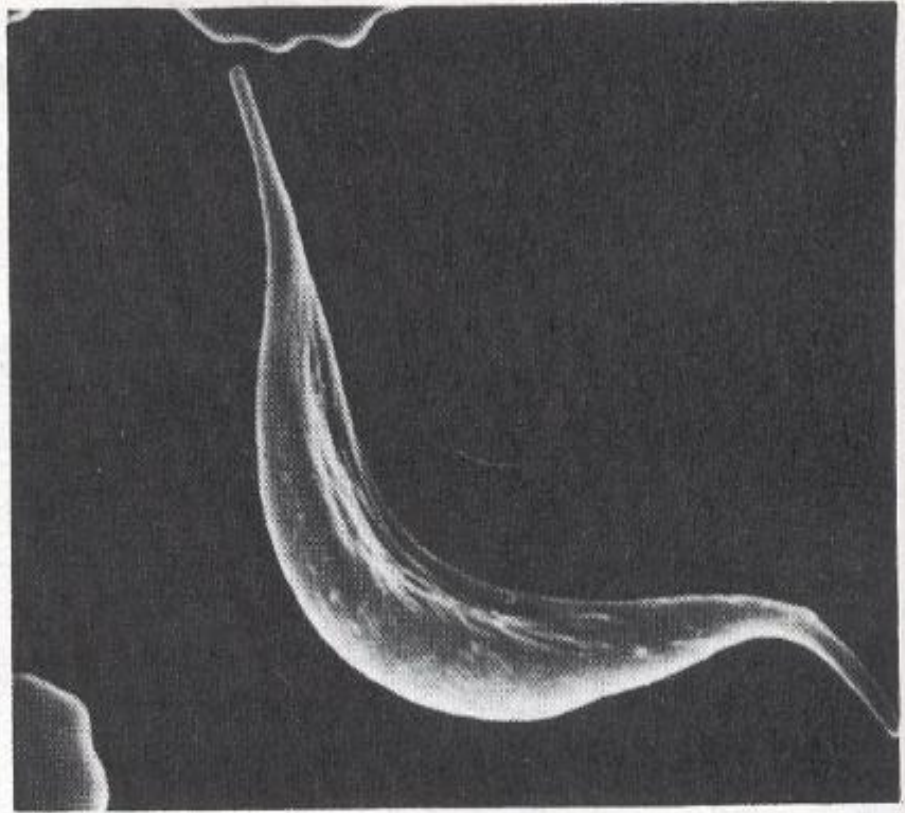



Biochemistry and Medicine:

- ◎ Biochemistry helped in elucidation of molecular causes of thousands of diseases; sickle cell disease, inborn errors of metabolism...**
- ◎ Nutrition and balanced diet**
- ◎ Disease diagnosis; scurvy deficiency of vitamin C**

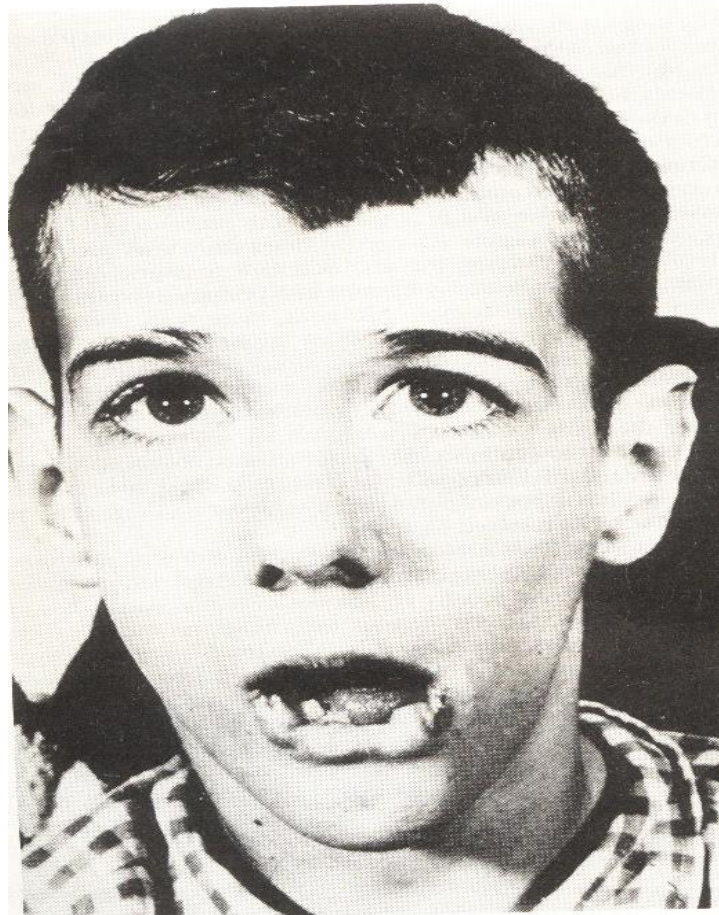


A



B

- The pt. present with mental retardation, spastic paraplegia, choreoathetosis and self-mutilation(Lesch Nyhan syndrome)



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- The Pt present with Mental retardation, Spastic paraplegia, choreoathetosis and self – Mutilation.



Water

- **is a molecule essential to life**
- **it solubilizes most biological molecules & affects most molecular interactions in biological systems & therefore modifies the properties of these molecules.**

Water

is :

- * product of oxidative metabolism of food**
- * reactant in many metabolic reactions**

Homeostasis

maintenance of composition of internal environment involves :

- * Distribution of water in the body**
- * Appropriate pH & electrolyte conc.**

Approx. 55 – 60 % of body weight is water

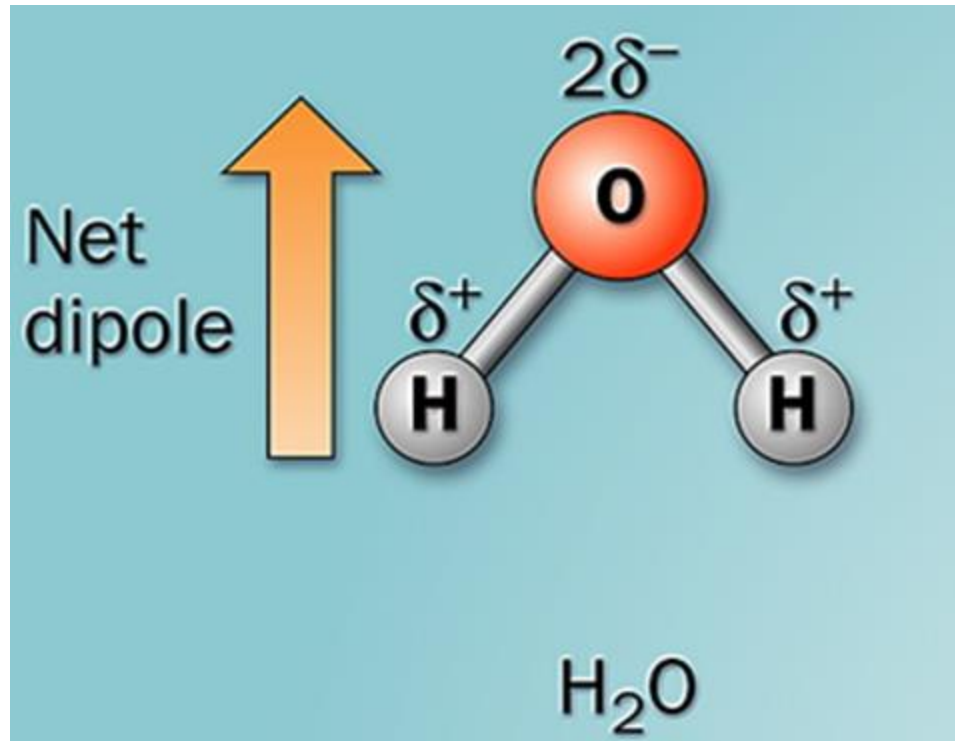
2/3 is intracellular fluid

1/3 is extracellular (25 % of which is plasma)

Water

- **Two properties of water are important to its function in biological systems :**
 - 1. It is a polar molecule – triangular, asymmetry in distribution of charge**

Water



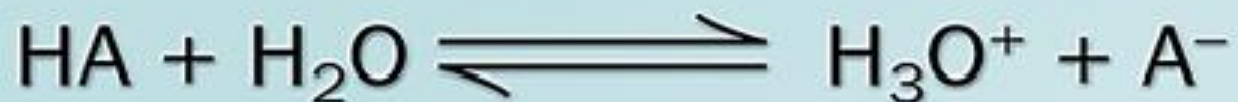
Water

- 2. It tends to associate – dipoles form H-bonds.**
- Water solvates polar molecules due to H-bond interactions & therefore it weakens H-bonds & electrostatic bonds between polar molecules**

Acids & Bases

An acid is a proton donor & a base is a proton acceptor.

Brønsted equation

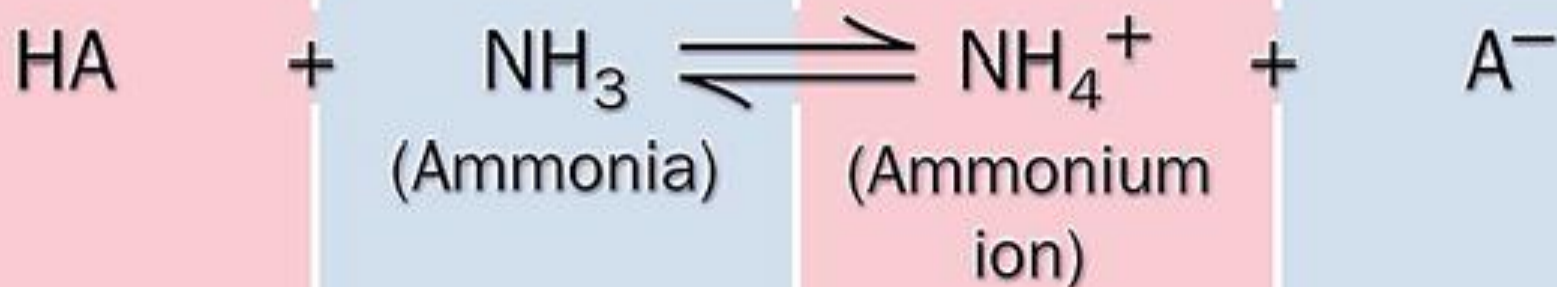
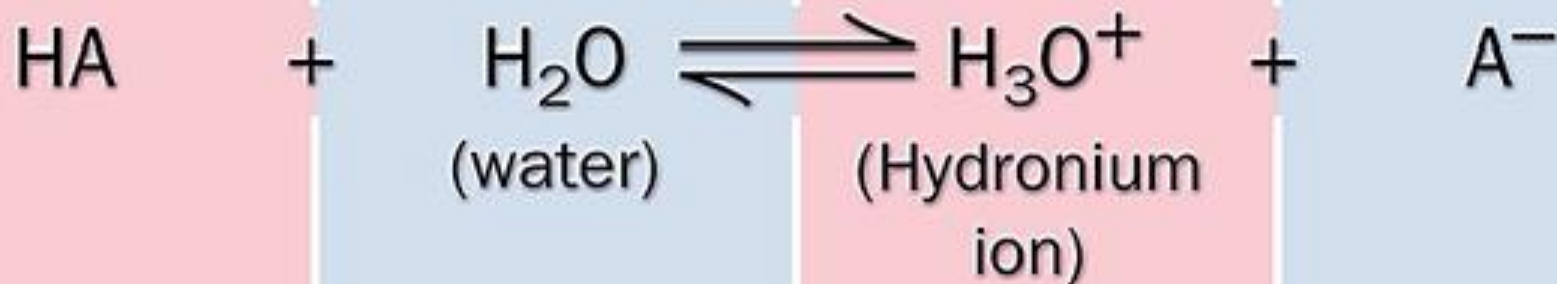


**Acid
(proton
donor)**

**Base
(proton
acceptor)**

**Conjugate
acid**

**Conjugate
base**



Strong & Weak Electrolytes

- Strong electrolytes ionize completely in solution.
- Weak electrolytes ionize only partially.
- Water is a weak electrolyte.
- $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$

pH

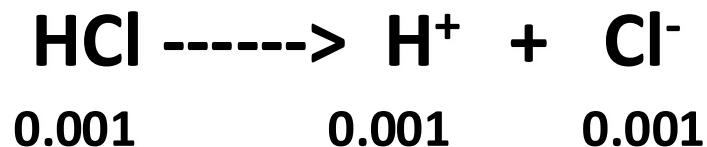
- To express very low values of $[H^+]$ in biological systems , the concept of pH was introduced by Sorensen :

pH is the negative $\log_{10}$ of hydrogen ion conc.

$$pH = -\log_{10} [H^+]$$

pH

Exp. What is the pH of 0.001 M HCl ?



$$[\text{H}^+] = 0.001 = 1 \times 10^{-3}$$

$$\text{pH} = -\log [\text{H}^+] = -(-3) = 3$$

pH of Weak acids and bases

- $\text{HA} \rightleftharpoons \text{A}^- + \text{H}^+$
- $K_a = [\text{H}^+][\text{A}^-]/[\text{HA}]$
- the Henderson Hasselbalch equation:
- $\text{pH} = \text{p}K_a + \log[\text{A}^-]/[\text{HA}]$

Dissociation constant pK_a

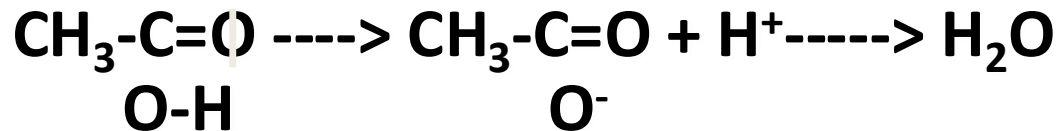
- the term pK_a is that pH at which an equivalent distribution of acid and conjugate base exists in solution.

Buffers

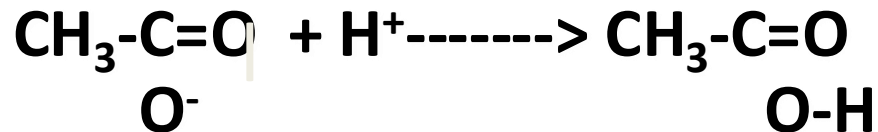
- **A buffer solution resists change in pH upon addition of acid or base.**
- **It consists of a mixture of a weak acid & its conjugate base e.g. acetic acid & potassium acetate.**

Buffers Mechanism of action

- * When base is added , it reacts with protons furnished by ionization of the weak acid.



- * When acid is added , it reacts with the acid anion(acetate) to give unionized acid.



Buffers

The effectiveness of a buffer called buffer capacity depends on :

- * Buffer conc. / both components**
- * Ratio of conj. acid to conj. base—maximum buffering capacity at a ratio of one**
i.e. $\text{pH} = \text{pK}_a$

Physiological Buffers

- **Acid/base balance is one of the major problems in clinical practice.**
- **H⁺ conc. In plasma & extracellular fluid is 40 nmol/L i.e. pH 7.4 (normal range 7.35 --- 7.45)**
- **pH below 7.0 & above 7.8 can have serious consequences**

Physiological Buffers

- Bicarbonate/Carbonic acid extracellular (plasma)
- Haemoglobin – intracellular (RBC ^s)
- $\text{NH}_3 / \text{NH}_4^+$ in urine
- Lactic acid
- Acetoacetic acid

Physiological Buffers

- **Proteins** mainly intracellular due to high conc. within cells. Plasma proteins play limited role in blood.
- **Phosphate buffer** intracellular $\text{H}_2\text{PO}_4^- / \text{HPO}_4^{=}$
Phosphate secreted in urine takes up H^+ , thus buffers urine.

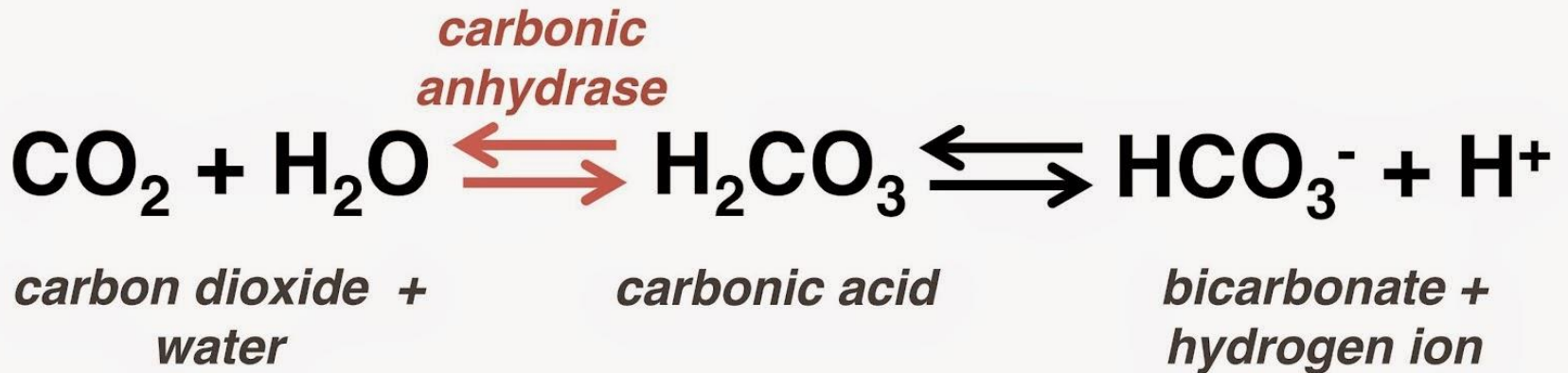
Regulation of blood pH

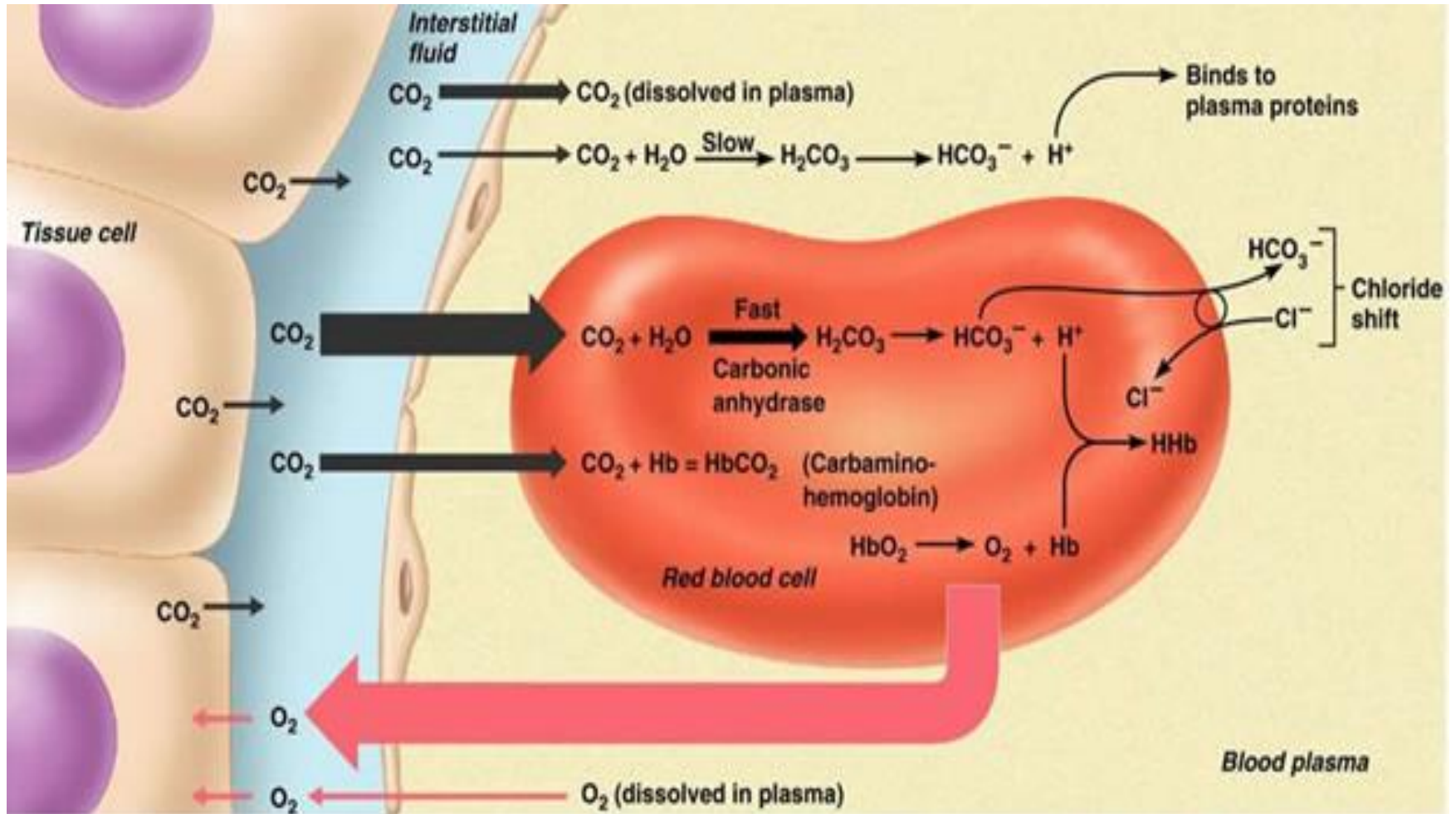
- is controlled by certain physiological mechanisms
 - * Blood buffers
 - * Respiratory mechanisms
 - * Renal mechanisms
- Buffering does not remove H^+ from the body ; only temporary short term solution -the body gets rid of H^+ by renal excretion

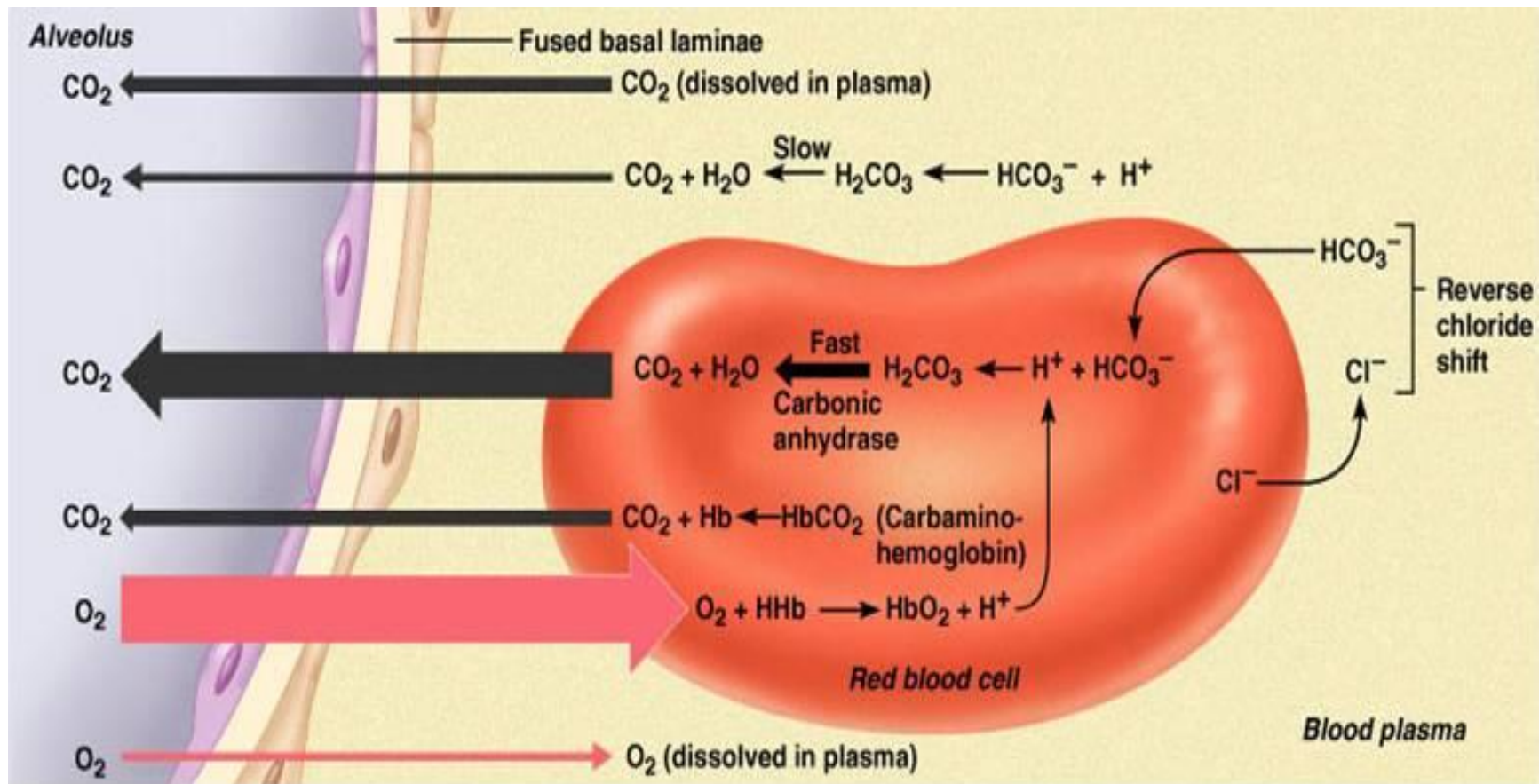
Bicarbonate/Carbonic acid buffer

- **Main ECF buffer system**
- **Buffering action & capacity due to the fact that carbon dioxide & carbonic acid are regulated by respiratory response ; furthermore bicarbonate is regulated by the kidneys.**

Bicarbonate/Carbonic acid buffer

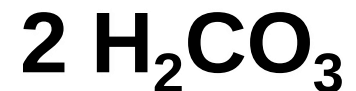
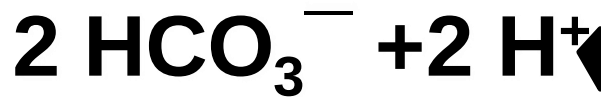
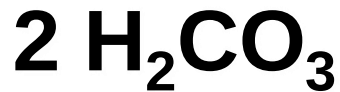
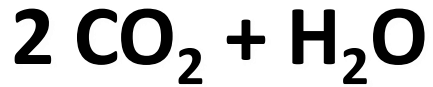






(b) Oxygen pickup and carbon dioxide release in the lungs

Exhaled



C.A.

Peripheral tissue

Lungs

Blood pH Levels



ACIDOSIS AND ALKALOSIS

Acidosis: $\text{pH} < 7.4$

Alkalosis: $\text{pH} > 7.4$

Respiratory acidosis: $\text{pCO}_2 \uparrow$

Respiratory alkalosis: $\text{pCO}_2 \downarrow$

Metabolic acidosis: $[\text{HCO}_3^-] \downarrow$

Metabolic alkalosis: $[\text{HCO}_3^-] \uparrow$

CAUSE**RESULT****COMPENSATION**

Respiratory acidosis	$p\text{CO}_2 \uparrow$	$[\text{H}^+] \uparrow$	$[\text{HCO}_3^-] \uparrow$	Metabolic alkalosis
Respiratory alkalosis	$p\text{CO}_2 \downarrow$	$[\text{H}^+] \downarrow$	$[\text{HCO}_3^-] \downarrow$	Metabolic acidosis
Metabolic acidosis	$[\text{HCO}_3^-] \downarrow$	$[\text{H}^+] \uparrow$	$p\text{CO}_2 \downarrow$	Respiratory alkalosis
Metabolic alkalosis	$[\text{HCO}_3^-] \uparrow$	$[\text{H}^+] \downarrow$	$p\text{CO}_2 \uparrow$	Respiratory acidosis